

SIMATS ENGINEERING (Saveetha School of Engineering)

Department of Artificial Intelligence and Data Science

ARTIFICIAL INTELLIGENCE REPORT

Experiment: Assessment Tool 1

Course Name: Artificial Intelligence

Course Code: CSA1704

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Aim

To develop a menu-driven Python program that demonstrates basic Artificial Intelligence concepts by implementing the Water Jug Problem, Mars Rover Intelligent Agent, 8-Queens Problem, OLA Cab Booking Goal-Based Agent, and Uniform Cost Search (UCS) algorithm.

Objectives

- To understand AI problem-solving techniques.
- To implement state-space search problems.
- To study intelligent agents and their characteristics.
- To implement uninformed search algorithms.
- To solve AI problems using Python programming.

Software Requirements

- Python 3.x
- IDLE / VS Code / PyCharm
- Windows 10/11

Theory

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, and problem-solving. AI uses intelligent agents that perceive their environment through sensors and act upon it using actuators to achieve specific goals.

In this experiment, five AI concepts are demonstrated:

1. **Water Jug Problem** – A state-space search problem solved by performing valid operations on two water jugs to reach the desired goal state.
 2. **Mars Rover Agent** – An intelligent autonomous agent that explores Mars, collects samples, avoids obstacles, and transmits scientific data.
 3. **8-Queens Problem** – A classical search problem solved using Backtracking and Depth-First Search (DFS) to place eight queens on a chessboard without conflicts.
 4. **OLA Cab Booking** – A Goal-Based Agent that selects and books an appropriate cab based on user input.
 5. **Uniform Cost Search (UCS)** – An uninformed search algorithm that always expands the node with the minimum cumulative path cost to find the least-cost path.
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Algorithm

1. Water Jug Problem

1. Start with both jugs empty.
 2. Fill the 3-gallon jug.
 3. Pour water into the 4-gallon jug.
 4. Repeat the process until exactly 2 gallons remain.
 5. Stop when the goal state is reached.
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2. Mars Rover Agent

1. Receive percepts from sensors.
 2. Analyze environmental conditions.
 3. Decide the appropriate action.
 4. Execute the action.
 5. Continue until the mission is completed.
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3. 8-Queens Problem

1. Begin with an empty chessboard.
 2. Place one queen in each column.
 3. Check for row and diagonal conflicts.
 4. If a conflict occurs, backtrack.
 5. Continue until all eight queens are placed safely.
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4. OLA Cab Booking

1. Read pickup and destination locations.
 2. Display available cab types.
 3. Allow the user to select a cab.
 4. Confirm booking.
 5. Assign a driver and start the trip.
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5. Uniform Cost Search

1. Insert the start node into the priority queue.
 2. Remove the node with the lowest cumulative cost.
 3. Expand neighboring nodes.
 4. Update cumulative costs.
 5. Repeat until the goal node is reached.
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Sample Output

Menu

===== ARTIFICIAL INTELLIGENCE =====

1. Water Jug Problem
2. Mars Rover Agent
3. 8 Queens Problem
4. OLA Cab Booking Agent
5. Uniform Cost Search (UCS)
6. Exit

Water Jug Problem

Goal Achieved!

4-Gallon Jug contains exactly 2 Gallons.

Mars Rover

Suitable Agent : Utility-Based Learning Agent

8 Queens

8 Queens Solution Displayed Successfully

OLA Cab Booking

Cab Booked Successfully

Driver Assigned

Happy Journey!

Uniform Cost Search

Least Cost Path : S -> B -> C -> G

Total Cost : 8

Result

The menu-driven Python program was successfully developed and executed. The program demonstrated the implementation of the Water Jug Problem, Mars Rover Intelligent Agent, 8-Queens Problem, OLA Cab Booking Goal-Based Agent, and Uniform Cost Search algorithm. All problems produced the expected outputs successfully.

Applications

- Robot Navigation
- Autonomous Vehicles
- Space Exploration
- Logistics and Route Optimization
- Intelligent Transportation Systems
- Game Playing
- AI-Based Decision Support Systems
- Resource Allocation

Advantages

- Easy to understand and implement.
- Demonstrates multiple AI concepts in a single program.
- Improves problem-solving and search strategy skills.
- Provides practical knowledge of intelligent agents.
- Beginner-friendly and suitable for academic learning.

Conclusion

The experiment successfully demonstrated the implementation of five fundamental Artificial Intelligence concepts using Python. The Water Jug Problem illustrated state-space search, the Mars Rover represented an intelligent agent, the 8-Queens Problem showcased backtracking search, the OLA Cab Booking system demonstrated a Goal-Based Agent, and Uniform Cost Search identified the least-cost path in a weighted graph. These implementations provide a strong foundation for understanding AI problem-solving techniques and intelligent agent design.